Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/30/2012
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	05/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Expedition 19/20 NOTE: End date changed to 5/31/2013, per NSSC (Ed., 6/6/12) NOTE: End date is now 5/31/2012 (from 5/31/2010) per NSSC information and F. Szofran/MSFC (10/2009) NOTE: Changed to FLIGHT project, per F. Szofran/MSFC (10/2009) NOTE: End date changed to 5/31/2010 (from 5/31/2009) per PI (12/08)		
Key Personnel Changes/Previous PI:	Nick Mauro has finished his PhD and is now working on an NSF-MRI grant in my research group. Two new students, Matt Blodgett and Jennifer Gewin, have moved onto this NASA project. The contribution of Dr. Anup Gangopadhyay to this project has been increased.		
COI Name (Institution):	Herlach, Dieter (Inst. fur Raumsimulation) Holland-Moritz, Dirk (Inst. fur Raumsimulation) Hyers, Robert (University of Massachusetts)		
Grant/Contract No.:	NNX07AK27G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Supercooling, the ability of a liquid held at temperatures below its melting temperature without crystallizing, was first discovered by Fahrenheit in 1724 when he was studying the freezing of water. It wasn't until 1952 that supercooling in liquid metals was demonstrated by David Turnbull. That liquids can be supercooled indicates that there is a barrier to the crystallization of the liquid; this is called the nucleation barrier. Nucleation is the first step in most first-order phase transitions, like the freezing of water. Nucleation is important in processes ranging from the crystallization of liquids to the development of bubbles in carbonated beverages to biological processes. It is even believed to be central in phase transitions that occurred in the early universe. An ability to control nucleation would be valuable for the development of new materials for advanced applications. In almost all cases nucleation data are described by the classical theory of nucleation, first developed in the late 19th century. It is now recognized however that there are serious problems with aspects of that theory in some regimes of application. Also, it does not properly describe nucleation when the chemical concentrations of the initial and final phases are different. Finally, recent evidence indicates that nucleation is often coupled to other processes and phase transitions, something that cannot be described readily within the classical theory. The task of the continuing ground-based research on this contract is the investigation of a coupling between structural ordering in the liquid and associated thermophysical properties, and with the nucleation barrier, and with other phase transitions that are being discovered that occur in supercooled liquids. Studies are focused on transition metals and their alloys. Ground-based measurements of the structure, specific heat, density, viscosity, surface tension, and emissivity in equilibrium and supercooled liquids are made using electrostatic levitation containerless processing techniques. In cooperation with ESA these data will be combined with studies made on the International Space Station to develop a better understanding of the complicated nucleation processes, and other phase transitions, that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The WU-BESL (Washington University Beamline Electrostatic Levitation) technique, developed under this contract, allows a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions. The discovery of possible new phase transitions in the liquid will lead to a better understanding of glass formation. The modification of WU-BESL to also include density, viscosity and surface tension measurements, in addition to structural data from synchrotron X-ray diffraction measurements, is allowing novel investigations of supercooled metallic liquids that is leading to a better understanding of how liquids order and how this relates to glass formation.
Task Progress:	Over the past year we have continued to refine the Washington University Beamline Electro-Static Levitation facility (WU-BESL) to enable thermophysical property measurements (density, viscosity, surface tension), in addition to the x-ray synchrotron studies for which WU-BESL was designed. We have used these expanded capabilities to better define the scientific questions that will be addressed in our upcoming ISS experiments, as well as to provide data needed for the definition of those experiments in terms of expected coil deposition in the EML and dust production. We have also modified our arc-melting facility to improve the quality of the vacuum in the arc-melting chamber in order to minimize oxygen contamination, which can profoundly influence the attainable supercooling in the alloy systems of interest. A brief description of the key results obtained over the past year follows: 1. Under partial support by this grant, in August 2010, we collected X-ray scattering data for 73 different alloy liquids at the Advanced Photon Source. The liquids were processed in a containerless environment using the WU-BESL, an electrostatic levitation facility that was designed and constructed at Washington University. The data collected provides support for thermophysical property measurements to be made on the ISS on our samples and on samples from other scientific teams. The structural information that can be gained from the scattering data can be used to understand features in the viscosity and specific heat measured on the ISS, which may be attributable to ordering and chemical phase transitions in the supercooled liquids. Reverse Monte Carlo (RMC) methods are often used to construct reasonable structural models for the liquid by fitting to the total structure factor, S(q) and the associated pair distribution function, g(r), which can be obtained from the scattered intensity data. To obtain the most accurate information, these fits should be constrained by results from theoretical ab-initio molecular dynamics (MD) simulations. However, in many cases such MD data are not available. We have applied a technique that has been successfully used in metallic glasses to determine the chemically specific partial pair distributions by simultaneously fitting g(r) data for a variety of liquids within the same family, but having different chemical compositions. This has been tested on the Cu-Zr liquids, which show evidence of chemical ordering and which are of interest to other teams within the Internal Working Group for the planned ISS experiments. The partials pair correlations obtained by this new method can be used to constrain RMC fits to provide more realistic information on chemical and topological ordering as a function of temperature. This information will enable more detailed studies of the liquids of interest. 2. To produce accurate values for S(q) and g(r), the measured X-ray scattering data must be corrected for the sample absorption of the incident and X-ray scattered intensity. While corrections exist for planar and cylindrical sample geometries, none exist for the spherical geometry of the liquid samples used in WU-BESL. This is a particularly important problem when the X-ray beam size is smaller than the sample diameter, which is the case here. The smaller beam and variations in sample size and density can cause positional variations between measurements, which can lead to off-axis transmission through the spherical sample and result in asymmetric scattering. We have derived an appropriate correction factor for this case. Tests on solid spherical samples of Vit 106A, a bulk metallic glass, demonstrate that predications from this new correction factor quantitatively reproduce experimental scattering data. This advance will improve the quality of the data obtained for scattering experiments that are planned for ISS samples. It goes further, however, providing a new tool for x-ray scattering studies made by others on spherical amorphous samples. 3. We have incorporated hardware to enable viscosity and surface tension measurements to be made in the WU-BESL. The data are obtained from levitated liquid samples by varying the vertical electric field at a regular rate, which sets up sample surface oscillations. The resonant frequency of these oscillations can be used to determine the surface tension. The rate at which they decay when the external excitation is removed can be related to the liquid viscosity. This new technique will allow ground-based studies on flight samples to better define the scientific objectives and procedures for studies on the ISS. It is currently being applied to the Cu-Zr liquids, to provide dynamical information that can be correlated with the structural data obtained from the X-ray scattering studies.

	4. We have made evaporation and dust production calculations for the Vit 106A sample that we will investigate on the ISS (as part of the THERMOLAB team) and for two Ti-Zr-Ni samples that were being considered as replacement samples for Batch 1 for the ISS experiments. The latter samples were moved into consideration when it became clear that the cup sample holders that were designed to minimize the impact of sample evaporation from high vapor pressure alloys did not survive the thermal shock test and would, therefore, not be available for the first batch. These calculations required experimental data that did not previously exist, and which was obtained using the WU-BESL. Calculations based on these experimental data demonstrated that the Vit 106A samples have minimal evaporation, confirming that they can be used in a cage sample holder instead of the cup holder, and clearing them for flight. The Ti-Zr-Ni samples similarly have low vapor pressures, but require parabolic flight studies before they can be flown on the ISS. Our studies suggested another substitute, Vit 106, which is closely related to Vit 106A, but which has interestingly different thermophysical properties. That alloy has now been chosen as an additional sample for Batch 1. Our ground-based work continues to lay a solid foundation for investigations of possible liquid/liquid and liquid/crystal phase transitions in supercooled liquids, providing complementary structural and thermophysical property information. Additionally, we are developing new tools that will have wide applicability for structural studies of amorphous systems, such as glasses and liquids. The results of these investigations, therefore, have both practical and basic interest.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
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